Software Engineering Lab

UE24CS341A

Lab 07: Fuzzing for Input Validation Bugs

SELF STORAGE RENTAL PROCESSING APPLICATION

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1. Objective

The objective of this lab is to use the Hypothesis fuzzing library to identify input-validation bugs in a Python application and improve the application so that it handles valid and invalid inputs in a controlled manner. The Self Storage application contains three functions: parse_rental(), duration_average(), and payable_rent().

2. Initial Application Execution

The original application was executed using representative valid inputs to observe the expected behaviour of the three functions. The inputs used were medium:6 for rental details ; 3,6,12 for rental duration values ; 2400|200 for rent and discount.

- **parse_rental()** – parsed the unit and rental duration and returned ('Medium', 6.0).
- **duration_average()** – calculated the integer average of the supplied rental durations and returned 7.
- **payable_rent()** – calculated the payable rent after applying 18% tax and subtracting the discount, returning 2632.0.

```
C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing>python self_storage.py
==== Running parse_rental ====
Enter unit:months (e.g. medium:6): medium:6
Unit and Months: ('Medium', 6.0)

==== Running duration_average ====
Enter rental months as CSV (e.g. 3,6,12): 3,6,12
Average Duration: 7

==== Running payable_rent ====
Enter rent|discount (e.g. 2400|200): 2400|200
Payable Rent: 2632.0

C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing>
```

Figure 1. Execution of the original Self Storage application with sample inputs.

3. Hypothesis-Based Fuzz Testing

Hypothesis was used with pytest to generate a range of inputs automatically, including arbitrary text and non-string values such as None. The initial test run exposed failures in all three functions. The failures showed that the original implementation assumed that inputs always had the expected type, separator, and numeric format.

The generated examples included empty strings, missing or incomplete separators, and None. These inputs caused errors such as unpacking errors, numeric conversion errors, and AttributeError when the functions attempted to call split() on a non-string value.

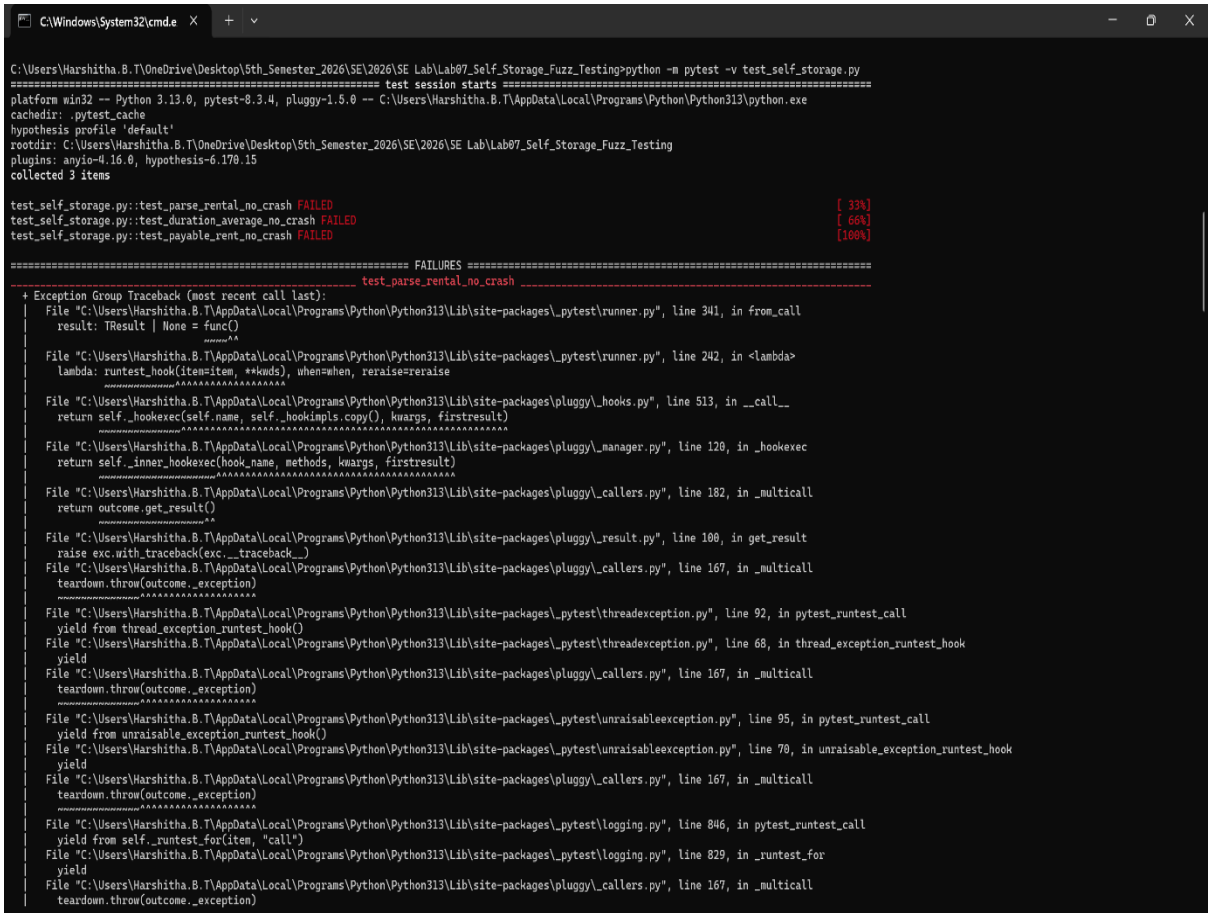
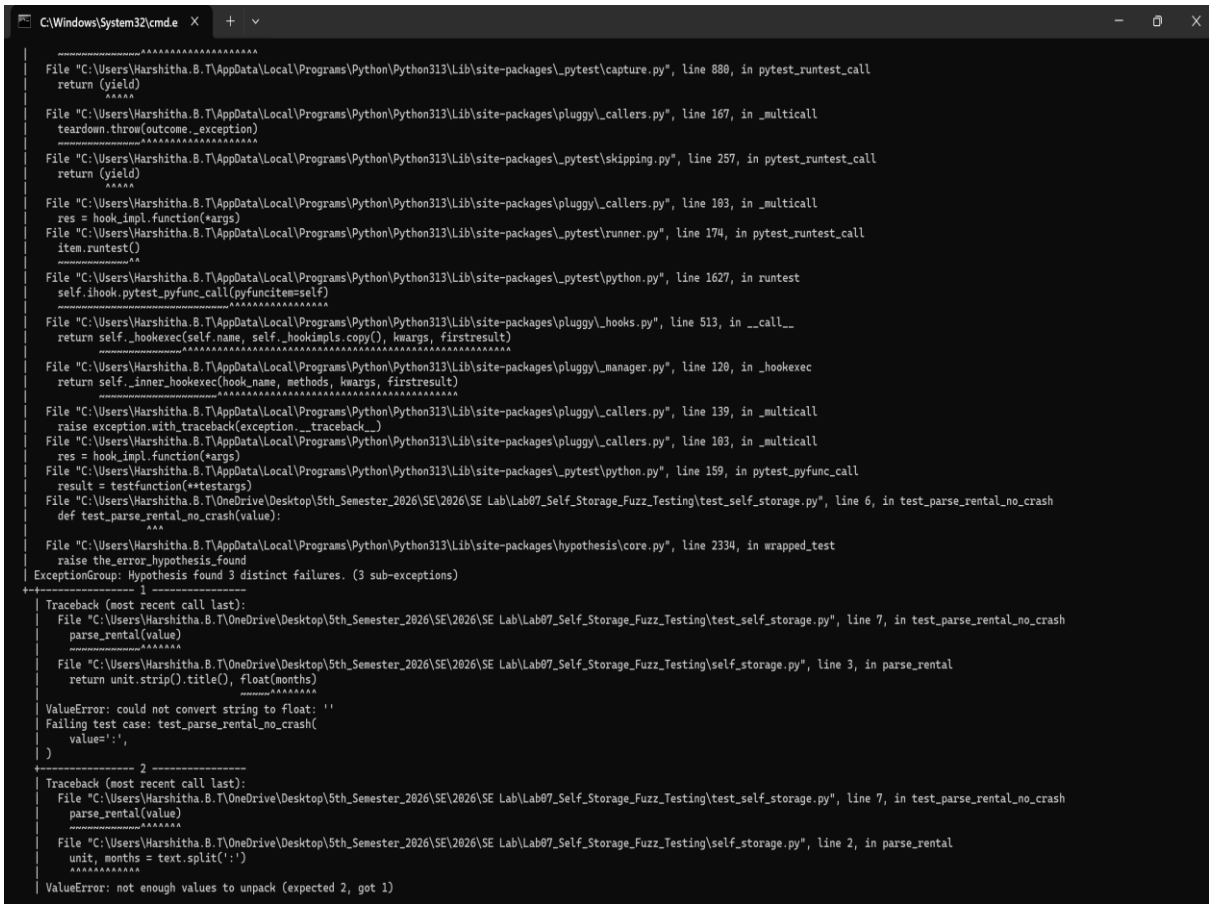


Figure 2A. Hypothesis output showing the failing test functions and generated failing examples.



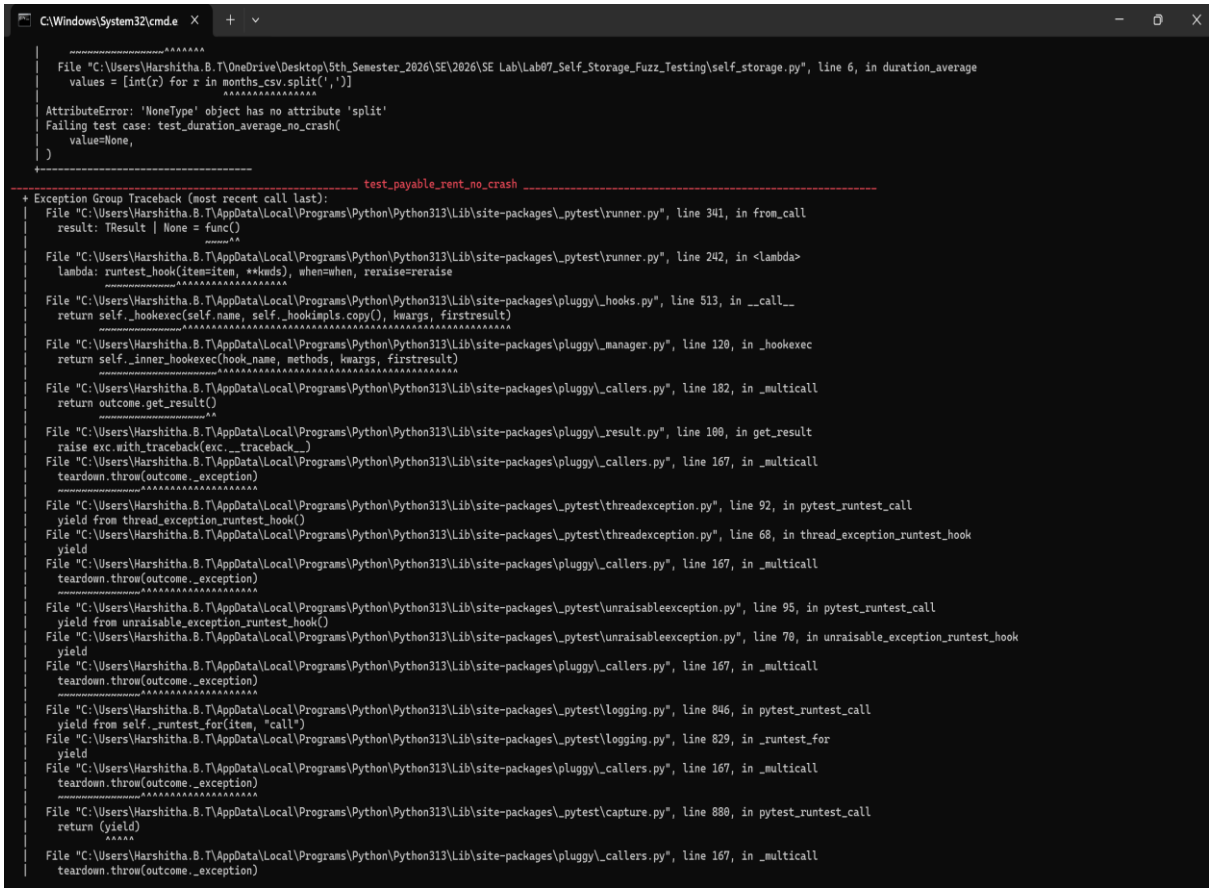
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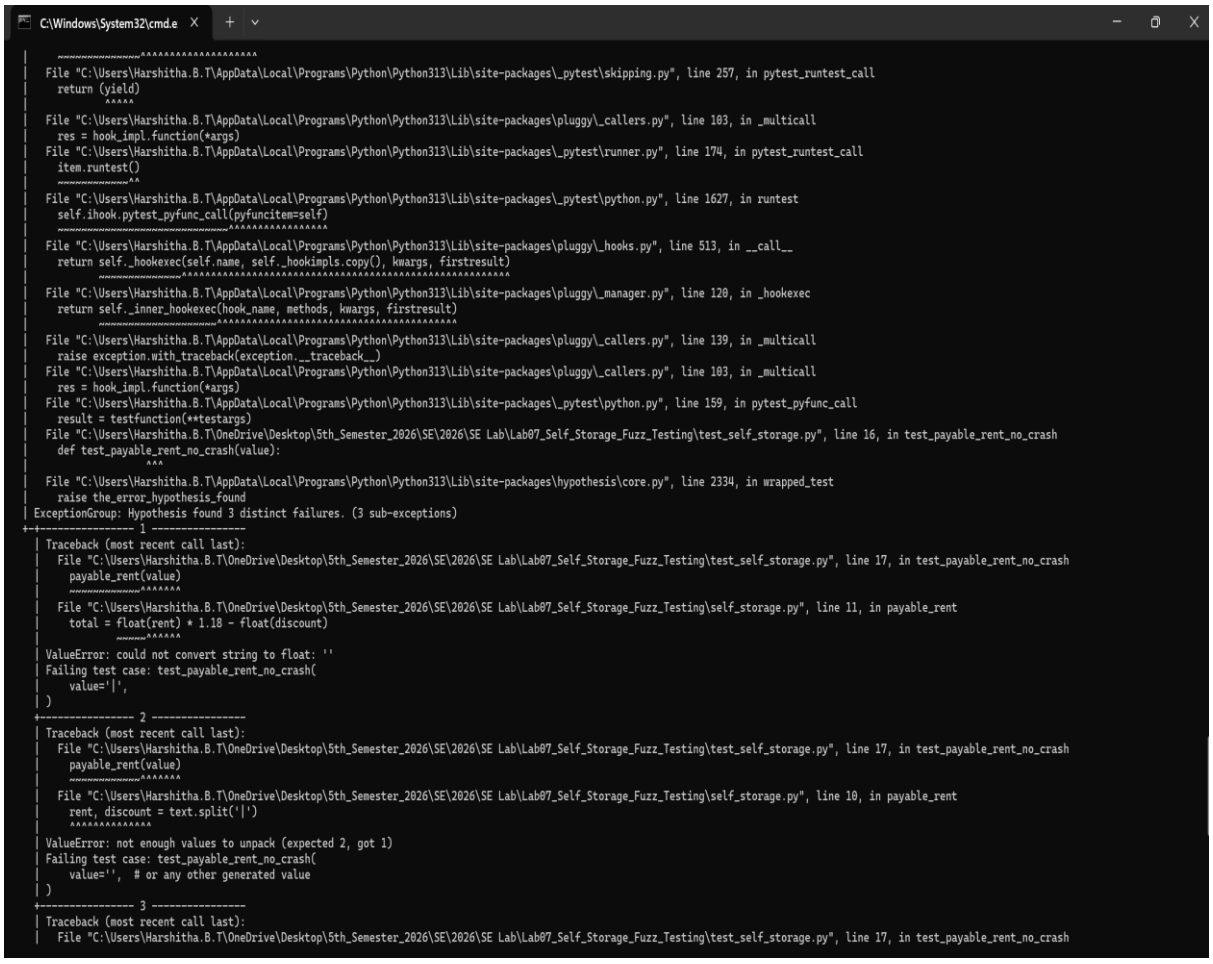
```
  | ValueError: not enough values to unpack (expected 2, got 1)
  | Failing test case: test_parse_rental_no_crash(
  |     value='',
  | )
  +---------------- 3 ----------------
  | Traceback (most recent call last):
  |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\test_self_storage.py", line 7, in test_parse_rental_no_crash
  |     parse_rental(value)
  |     ^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\self_storage.py", line 2, in parse_rental
  |     unit, months = text.split(':')
  |                    ^^^^^^^^^^
  | AttributeError: 'NoneType' object has no attribute 'split'
  | Failing test case: test_parse_rental_no_crash(
  |     value=None,  # or any other generated value
  | )
  +------------------------------------
_____________________________________________________ test_duration_average_no_crash _____________________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                             ^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
```

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```
  |     yield
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  |     item.runtest()
  |     ^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 1627, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |     ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\test_self_storage.py", line 11, in test_duration_average_no_crash
  |     def test_duration_average_no_crash(value):
  |                ^^^
  |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\test_self_storage.py", line 12, in test_duration_average_no_crash
    |     duration_average(value)
    |     ^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\self_storage.py", line 6, in duration_average
    |     values = [int(r) for r in months_csv.split(',')]
    |                  ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_duration_average_no_crash(
    |     value='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\test_self_storage.py", line 12, in test_duration_average_no_crash
    |     duration_average(value)
    |     ^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\self_storage.py", line 6, in duration_average
```



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    |     ~~~~~~~~~~~~~~~~~^^^^^^^
    |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\self_storage.py", line 6, in duration_average
    |     values = [int(r) for r in months_csv.split(',')]
    |                              ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_duration_average_no_crash(
    |     value=None,
    | )
    +------------------------------------
_____________________________________________ test_payable_rent_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
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    |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
    |             ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
    |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
    |            ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
    |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
    |            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
    |     return outcome.get_result()
    |            ~~~~~~~~~~~~~~~~~^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_result.py", line 100, in get_result
    |     raise exc.with_traceback(exc.__traceback__)
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    |     yield
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
    |     teardown.throw(outcome._exception)
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    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
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    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
    |     yield
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
    |     teardown.throw(outcome._exception)
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    |     yield
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    |     teardown.throw(outcome._exception)
    |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
    |     return (yield)
    |             ^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
    |     teardown.throw(outcome._exception)
```



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    |     ~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
    |     return (yield)
    |             ^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
    |     res = hook_impl.function(*args)
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
    |     item.runtest()
    |     ~~~~~~~~~~~~^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 1627, in runtest
    |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
    |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
    |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
    |            ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
    |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
    |            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
    |     raise exception.with_traceback(exception.__traceback__)
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
    |     res = hook_impl.function(*args)
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
    |     result = testfunction(**testargs)
    |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\test_self_storage.py", line 16, in test_payable_rent_no_crash
    |     def test_payable_rent_no_crash(value):
    |                ^^^
    |   File "C:\Users\Harshitha.B.T\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
    |     raise the_error_hypothesis_found
    | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
    +-+---------------- 1 ----------------
      | Traceback (most recent call last):
      |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\test_self_storage.py", line 17, in test_payable_rent_no_crash
      |     payable_rent(value)
      |     ~~~~~~~~~~~~^^^^^^^
      |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\self_storage.py", line 11, in payable_rent
      |     total = float(rent) * 1.18 - float(discount)
      |             ~~~~~~~~~~~~^^^^^^
      | ValueError: could not convert string to float: ''
      | Failing test case: test_payable_rent_no_crash(
      |     value='|',
      | )
      +---------------- 2 ----------------
      | Traceback (most recent call last):
      |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\test_self_storage.py", line 17, in test_payable_rent_no_crash
      |     payable_rent(value)
      |     ~~~~~~~~~~~~^^^^^^^
      |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\self_storage.py", line 10, in payable_rent
      |     rent, discount = text.split('|')
      |     ^^^^^^^^^^^^^^
      | ValueError: not enough values to unpack (expected 2, got 1)
      | Failing test case: test_payable_rent_no_crash(
      |     value='',  # or any other generated value
      | )
      +---------------- 3 ----------------
      | Traceback (most recent call last):
      |   File "C:\Users\Harshitha.B.T\OneDrive\Desktop\5th_Semester_2026\SE\2026\SE Lab\Lab07_Self_Storage_Fuzz_Testing\test_self_storage.py", line 17, in test_payable_rent_no_crash
```

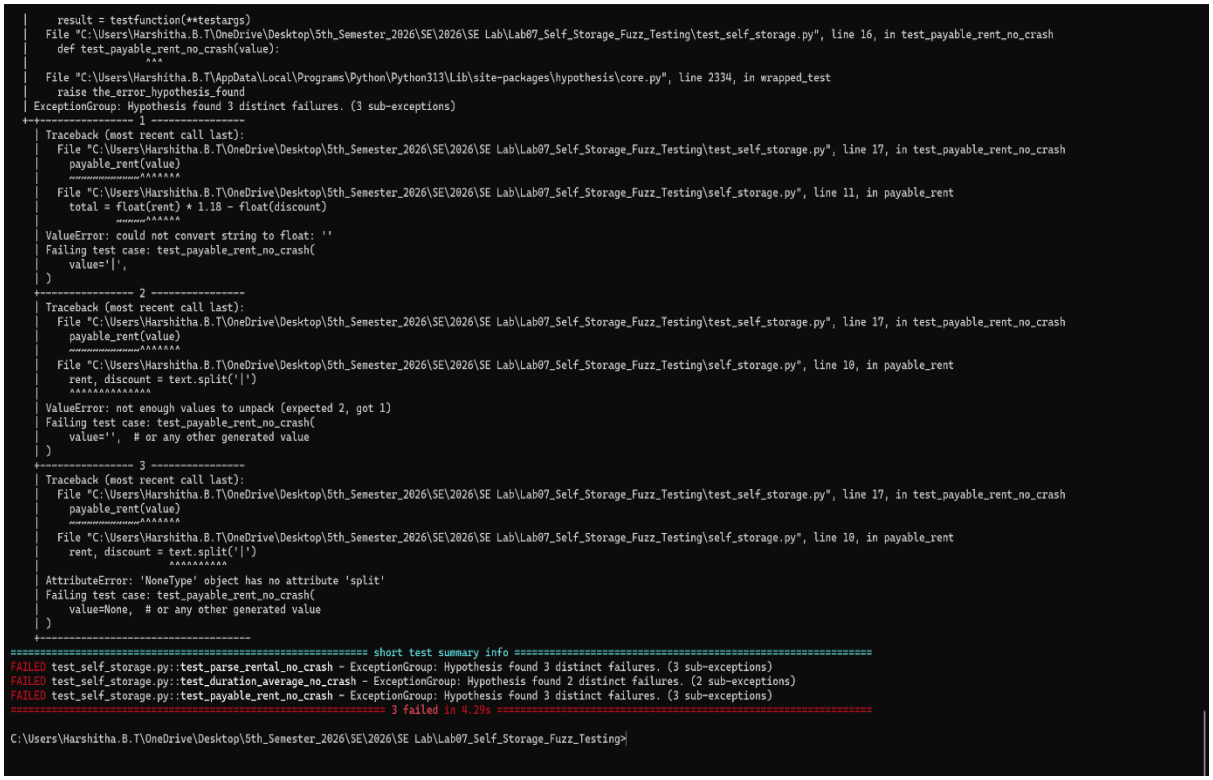


Figure 2B. Pytest summary showing that all three initial test functions failed.

4. Bug Identification and Application Fixes

The application was updated to validate inputs before performing parsing or calculations. Invalid inputs are rejected using clear ValueError exceptions instead of causing unexpected runtime errors.

- **parse_rental()** – checks that the input is text, contains exactly one colon, has a non-empty unit, and contains a finite, non-negative numeric duration.
- **duration_average()** – checks that the input is text and contains non-empty comma-separated integer values. Negative durations are rejected before calculating the average.
- **payable_rent()** – checks that the input is text, contains exactly one vertical-bar separator, and provides finite, non-negative numeric values for rent and discount.

The interactive application was also updated so that the corrected functions are used by the program when processing user input.

5. Execution of the Fixed Application

After applying the input-validation changes, the corrected application was executed again using the same representative inputs: medium:6, 3,6,12, and 2400|200. The application produced the expected results: ('Medium', 6.0), 7, and 2632.0 respectively.

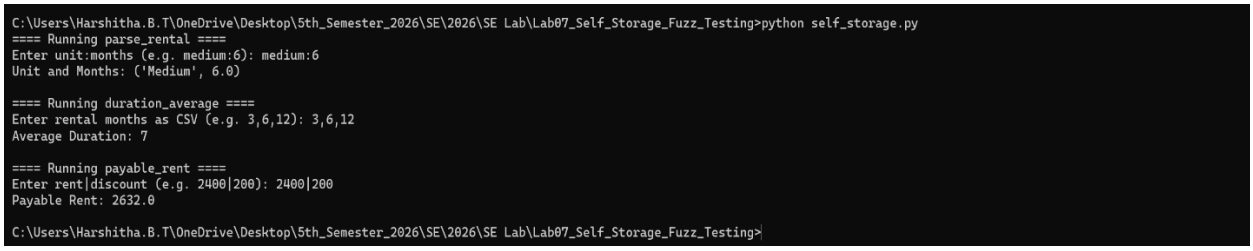


Figure 3. Execution of the corrected Self Storage application with the same sample inputs.

6. Testing After Fixing

The Hypothesis test suite was executed again against the corrected application. The suite contains six property-based tests covering generated valid inputs and arbitrary inputs for all three functions. The tests verified the expected calculation results for valid inputs and checked that invalid inputs are handled safely.

All six tests passed in the final test run, confirming that the corrected functions satisfy the properties checked by the test suite.

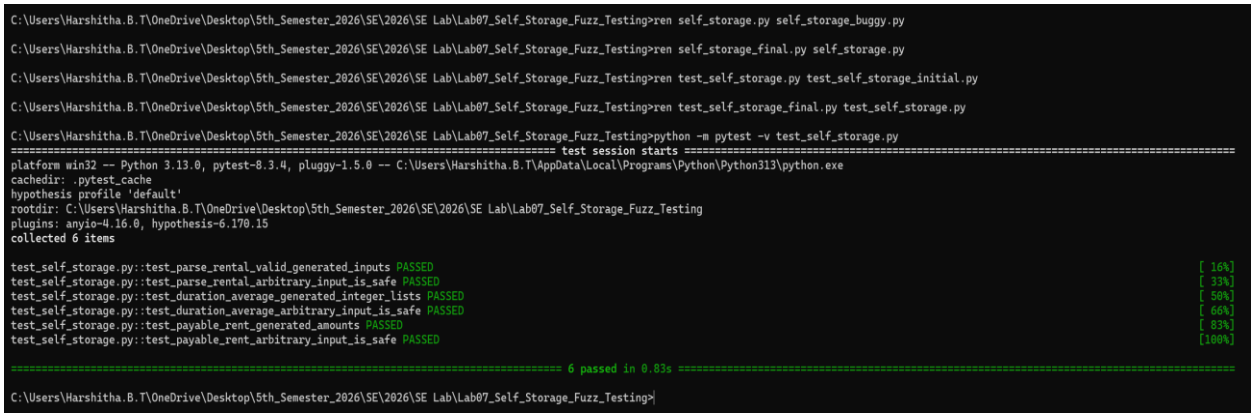


Figure 4. Final pytest output showing all six Hypothesis tests passing.

7. Conclusion

Hypothesis-based fuzz testing helped identify input-validation weaknesses in the original Self Storage application by generating edge cases that were not covered by the sample inputs. The application was updated to validate input types, separators, and numeric values before processing them. The final test run completed successfully with all six tests passing, demonstrating that the corrected application handles the tested valid and invalid inputs as expected.

~ Thank you ~